

1. Business Problem

Which three vendors should we put on PIP this quarter — and which vendors deserve more business?

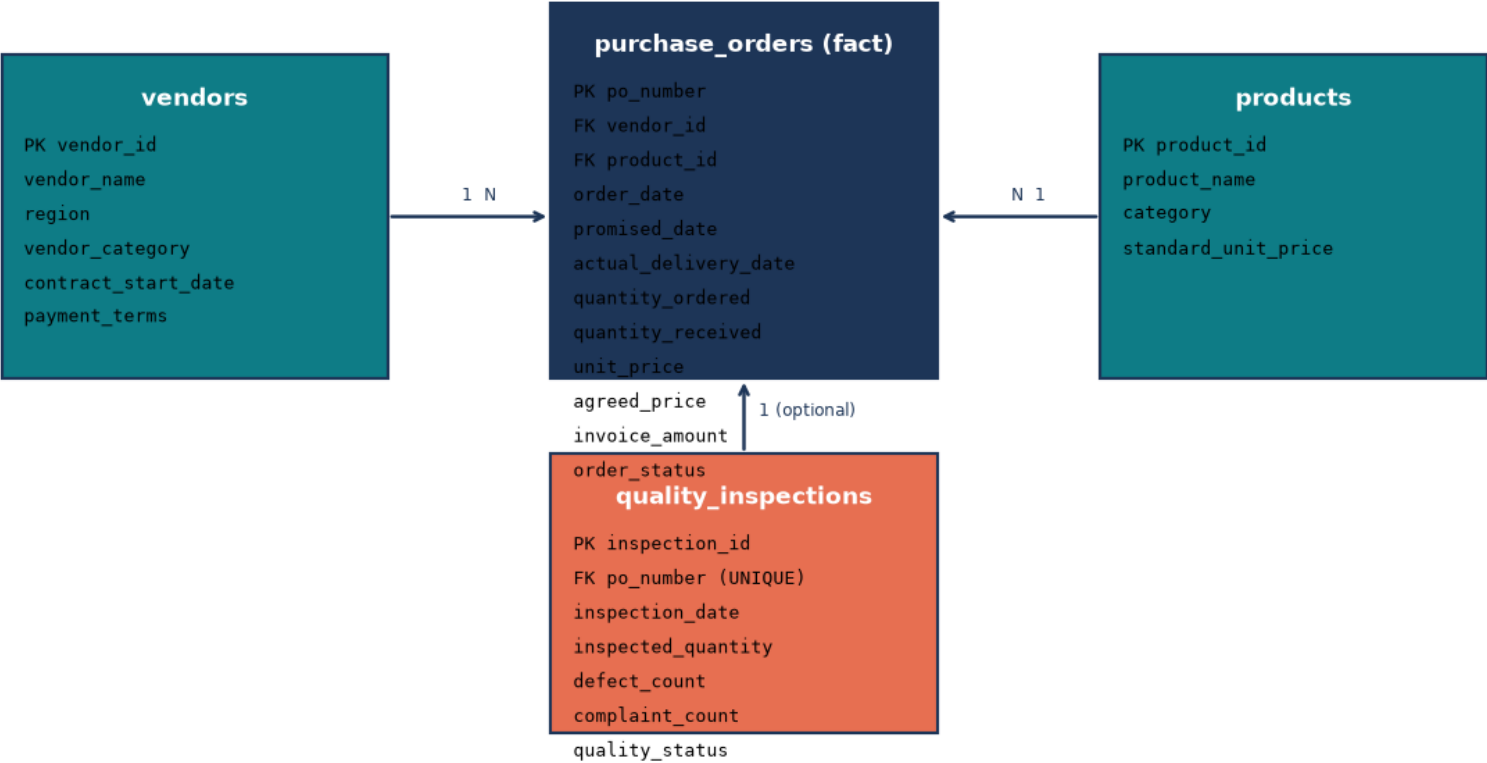
- NorthBridge Supplies procures from 20 vendors across 4 categories and 4 regions.
- Procurement leadership needs an evidence-based quarterly review — not gut feel.
- Report **supports** the decision; sole-source and total-cost checks stay with the category manager.

2. Data & Tools

~1,400 POs • Jan–Sep 2025 • 20 vendors • ■110.9 Cr spend

- 4-table star schema — vendors, products, purchase_orders (fact), quality_inspections (subset, ~60% coverage, UNIQUE po_number).
- Python (pandas + matplotlib) for cleaning & EDA — duplicates, casing, missing defects, over-shipments all handled with logged decisions.
- PostgreSQL 16 for KPI queries (CTEs, LEFT JOIN on inspections, window functions, LAG for MoM change).
- Power BI Desktop for the 2-page dashboard (mockups shipped as PNGs; DAX file included).

NorthBridge Supplies — ER diagram



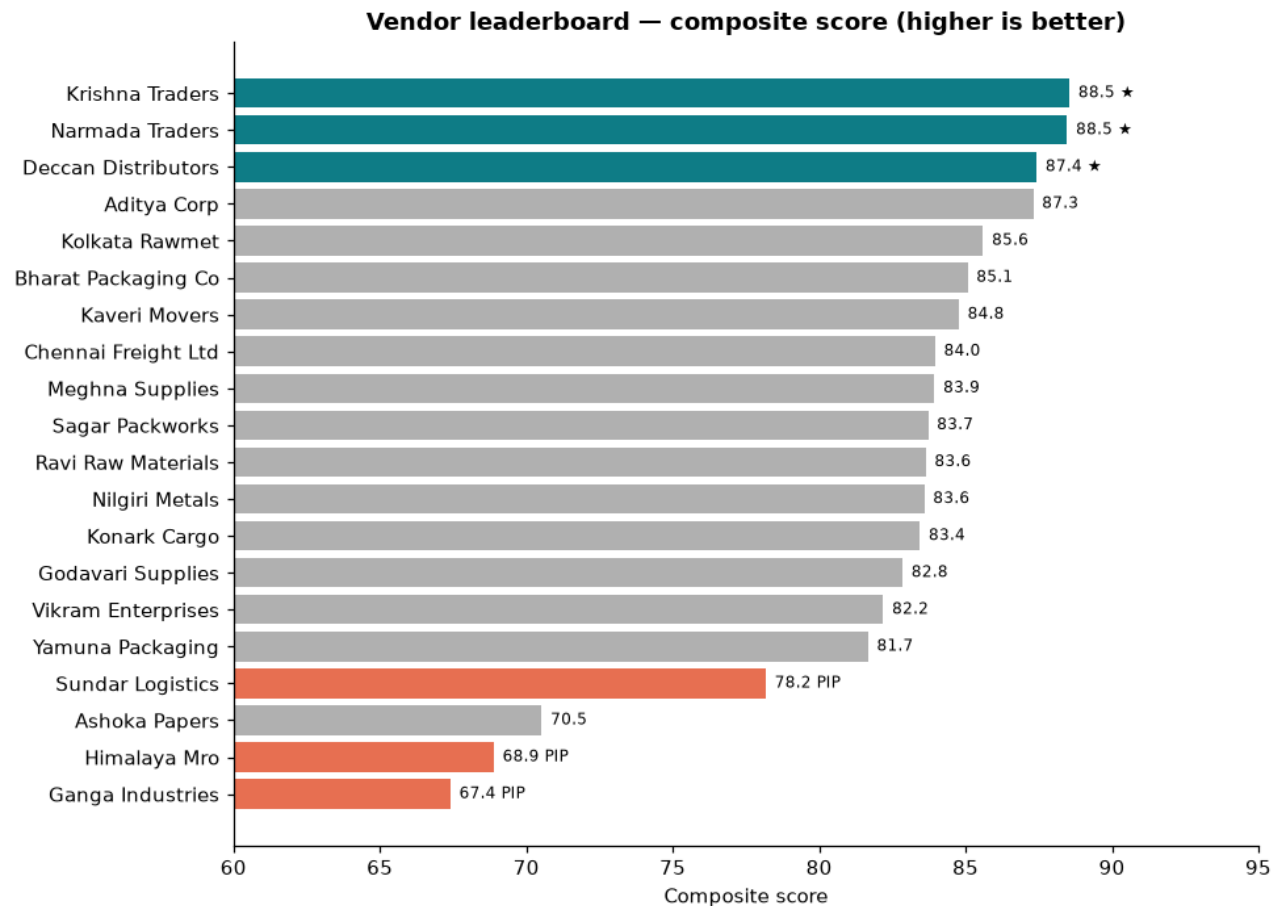
3. Analysis & KPIs

Five KPIs → one composite score, guarded by an eligibility rule

- **OTD %**, **Defect Rate %** (inspected only), **Price Variance %**, **Lead Time**, **Fill Rate %**.
- **Composite** = $0.4 \cdot \text{OTD} + 0.4 \cdot (100 - \text{Defect}) + 0.2 \cdot (100 - \max(0, \text{PriceVar}))$. Weights are academic — documented, not industry-standard.
- Vendors need ≥ 20 POs to be eligible → small-sample noise is filtered out (Ashoka Papers, 17 POs, is excluded even though its score is low).
- PIP filter: bottom-3 score AND below-avg OTD or above-avg defect rate. Prevents flagging a chronic-low-score vendor with no operational issue.

4. Dashboard & Findings

Executive Summary → Vendor Deep-Dive (drill-through)



- PIP shortlist: **Ganga Industries, Himalaya Mro, Sundar Logistics**. Reward shortlist: **Krishna Traders, Narmada Traders, Deccan Distributors**.
- Sundar Logistics defect rate jumps 1.9% → 10.3% after Month 4 — a directional issue an annual average would have hidden.

5. Recommendations & Learnings

Business action → data & analytics learnings

Business:

- Put **Ganga Industries, Himalaya Mro, Sundar Logistics** on a 90-day PIP; review after next quarter's data.
- Retain and consider expanding volumes with **Krishna Traders, Narmada Traders, Deccan Distributors** — pending sole-source and capacity checks with the category manager.
- Raise QC inspection coverage on vendors with low sample sizes before the next review.

Learnings:

- Averages can hide directional problems — always look at the trend.
- Explicit eligibility rules make a scoring system defensible.
- Documenting cleaning decisions is what separates an analyst from a spreadsheet.